Date: 2025-03-21
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兴业证券数据库应用 redo 卷时延增高问题分析报告

Product Type/Serial Number	Microcode Revision	Case Number
VSP F5100/31031&31029	90-09-22-00/00	05047439

目录

1 问题概述

从 2025/3/10 开始，有两套数据库应用（UF20-DB01BHP 和 ZYO32_DB01JQ）所使用的 redo 日志卷陆续出现时延增高的现象，统计到的事件如下（以 ZYO32_DB01JQ 为例）：

```
*** 2025-03-10 10:08:00.374
Warning: log write elapsed time 30839ms, size 7KB
*** 2025-03-10 10:08:24.394
--
*** 2025-03-10 10:19:05.334
Warning: log write elapsed time 30125ms, size 1KB
*** 2025-03-10 10:19:29.342
--
*** 2025-03-10 13:02:20.342
Warning: log write elapsed time 30420ms, size 1KB
*** 2025-03-10 13:03:35.341
--
*** 2025-03-10 13:09:10.534
Warning: log write elapsed time 30370ms, size 0KB
*** 2025-03-10 13:09:16.543
--
*** 2025-03-10 13:50:49.334
Warning: log write elapsed time 30155ms, size 2KB
*** 2025-03-10 13:51:01.341
--
*** 2025-03-10 13:57:14.359
Warning: log write elapsed time 30504ms, size 1KB
*** 2025-03-10 13:57:47.360
--
*** 2025-03-10 14:03:02.751
Warning: log write elapsed time 11065ms, size 1KB
*** 2025-03-10 14:03:11.755
--
*** 2025-03-10 14:28:47.574
Warning: log write elapsed time 30814ms, size 1KB
*** 2025-03-10 14:29:50.581
--
*** 2025-03-10 14:42:13.240
```

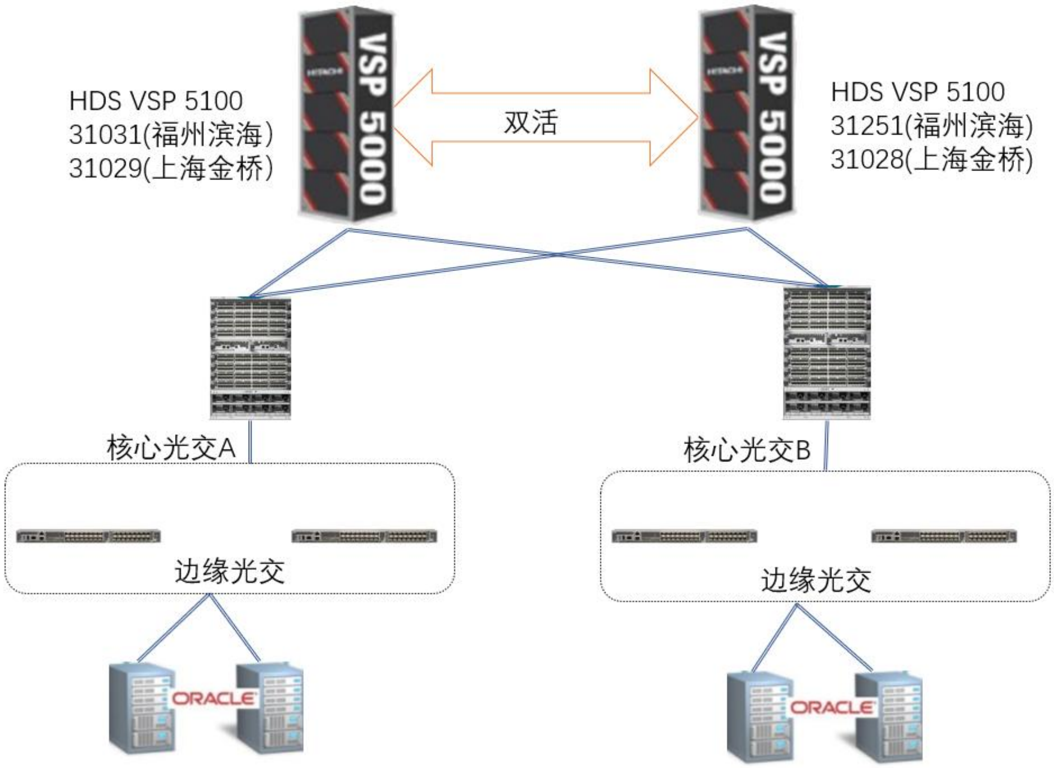
```
Warning: log write elapsed time 11178ms, size 2KB
NSS2 is not running anymore.
--
*** 2025-03-10 14:55:53.264
Warning: log write elapsed time 11115ms, size 4KB
*** 2025-03-10 14:55:56.284
--
*** 2025-03-11 09:41:35.222
Warning: log write elapsed time 11021ms, size 2KB
*** 2025-03-11 09:42:32.234
--
*** 2025-03-11 09:50:22.335
Warning: log write elapsed time 30971ms, size 6KB
*** 2025-03-11 09:50:28.340
--
*** 2025-03-11 10:06:32.714
Warning: log write elapsed time 11037ms, size 1KB
*** 2025-03-11 10:06:59.721
--
*** 2025-03-11 10:32:39.325
Warning: log write elapsed time 11227ms, size 8KB
*** 2025-03-11 10:33:21.331
--
*** 2025-03-11 13:14:10.358
Warning: log write elapsed time 30656ms, size 1KB
*** 2025-03-11 13:14:19.361
--
*** 2025-03-11 13:55:35.638
Warning: log write elapsed time 30959ms, size 1KB
*** 2025-03-11 13:56:08.641
--
*** 2025-03-11 19:55:31.390
Warning: log write elapsed time 11359ms, size 1KB
*** 2025-03-11 19:56:10.392

*** 2025-03-12 09:40:35.638
Warning: log write elapsed time 30239ms, size 1KB
*** 2025-03-12 09:44:38.642
--
*** 2025-03-12 10:34:49.422
Warning: log write elapsed time 11062ms, size 5KB
```

```
*** 2025-03-12 10:40:04.431
--
*** 2025-03-12 11:13:31.351
Warning: log write elapsed time 30087ms, size 3KB
*** 2025-03-12 11:13:49.368
```

这两套应用分别连接到位于福州滨海数据中心和上海金桥数据中心的日立 GAD 双活存储，涉及的这 4 台存储配置情况如下表。

	bh71	bh72	jq71	jq72
SN	31031	31251	31029	31028
安装位置	福州滨海	福州滨海	上海金桥	上海金桥
磁盘	50*7.6TB NVMe 6D+2P	50*7.6TB NVMe 6D+2P	50*7.6TB NVMe 6D+2P	50*7.6TB NVMe 6D+2P
可用容量 (TB)	247.58	247.58	247.58	247.58
Cache	1TB	1TB	1TB	1TB
CHB	4 对	4 对	4 对	4 对
前端口	24*32Gbps	24*32Gbps	24*32Gbps	24*32Gbps
Firmware	90-09-22-00/00	90-09-22-00/00	90-09-22-00/00	90-09-22-00/00
SVOS	9.9	9.9	9.9	9.9
备注	GAD Primary	GAD Secondary	GAD Primary	GAD Secondary

应用和存储的连接拓扑示意图如下：



应用使用的日志卷为：

- UF20-DB01BHP：03:22，使用的存储端口为：3C/4D
- ZYO32_DB01JQ：40:6B，使用的存储端口为：3A/4B

2 问题分析

2.1 存储 Dump 日志分析

2.1.1 应用对应的 MDKC 存储日志分析

从 MDKC 存储（31029）底层日志可以看到 ZYO32_DB01JQP 应用对应的 4B 端口在部分时段出现了 I/O 传输延迟的现象，出现延迟的卷正是这套应用所使用的 Redo 日志道卷，大部分为写 I/O（SCSI CMD=2A:write）方向，根据出现的 SSB 特征码 D034/D031，I/O 传输延迟的原因为“等待数据传输”，问题指向存储外部链路端，其构成组件包括

1)FC-HBA(WWN=100000620B3AC203)、2)FC-HBA 到交换机的光纤线、3)FC-HBA 在交换机上的 SFP 模块、4)存储 4B 端口上的 SFP 模块、5)存储 4B 到交换机的光纤线、6) 存储 4B 端口在交换机上的 SFP 模块：

```
2025/03/11 10:32:42   B6C0#  3C   4B          : xlmain It FCP Frame-receives from non-LOGIN HOST.
2025/03/11 10:32:42   B358#  3F   4B          : qlsnhlliocb LL IOCB was issued
2025/03/11 10:32:42   B3FF   3F   4B          : snptovssb Information enhancement for transfer TOV
2025/03/11 10:32:42   B6AD#  3F   4B          : tpcc_xlfhostssb Information enhancement for transfer TOV/count error (Host information: Port ID, WWN)
2025/03/11 10:32:42   B6A9#  3F   4B          : snptov   Complementing information log for transfer TOV (LM)
2025/03/11 10:32:42   D034#  3F   4B   04:6B  : fcmxrwtov Sync command Read/Write JOB TOV
2025/03/11 10:32:42   D031#  3F   4B   04:6B  : fcmxssbtv Additional information of SSB=d030/d034 (OPEN command, SCSI reset/TOV)
2025/03/11 10:32:42   DDA1#  3F   4B          : cktov_lm Data transfer (Tachyon) completion TOV
2025/03/11 10:32:42   DDA1#  3F   4B   04:6B  : cktov_lm Data transfer (Tachyon) completion TOV
2025/03/11 10:32:42   FFFB#       4B   LOGO   : LINK ELS EVENT Type = LOGO
2025/03/11 10:32:33   D555#  3F   4B   04:6B  : fcmxd555s Information enhancement log was output because job running time of more than 1 sec was detected
2025/03/11 09:11:28   10F5   07               : timdiv Async processing was performed forcibly
2025/03/11 08:54:30   10F5   0E               : timdiv Async processing was performed forcibly
2025/03/11 07:11:36   10F5   10               : timdiv Async processing was performed forcibly
2025/03/11 07:00:43   10F5   00               : timdiv Async processing was performed forcibly
```

Refcode = D034 : fcmxrwtov Sync command Read/Write JOB TOV

DATE TIME	PORT	LDEV#	CMD	KEY/ASC	HG-NAME	WAIT REASON	HOST WWN / IQN	FCID
2025/03/11 19:55:32	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/03/11 10:32:42	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/13 23:11:12	4B	00:04:6C	8A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/13 23:01:26	4B	00:04:6C	8A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/13 09:59:30	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/13 09:53:15	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/13 09:49:28	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000
2025/02/12 10:10:25	4B	00:04:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203	490000

根据存储端 4B 端口的错误计数器，没有发现用于判断 4B 端口 SFP 模块和 4B 端口到交换机的连线的计数器有异常增长情况，因此可以排除 4B 端口 SFP 模块和 4B 端口到交换机的连线问题：

- CRC: 如果该计数器不为 0，基本上为存储端口 SFP 模块问题；
- EOFa: 如果该计数器不为 0，基本上为存储端口到交换机之间的连线问题

```
PORT LABEL = 4B
16G FC Port Statistics
| Hex_Count | Decimal_Count | Port_Stats_Description
|-----------|---------------|------------------------------------------------------------------------------------------
| 00000000  |             0 |Link Failure
| 00000000  |             0 |Loss of Sync
| 00000000  |             0 |Loss of Signal
```

```
| 00000000  |           0  |Primitive Sequence Protocol Error
| 00000000  |           0  |Invalid Transmission Word
| 00000000  |           0  |Invalid CRC
| 00000000  |           0  |LIP Occurred
| 00000000  |           0  |Link UP
| 00000000  |           0  |Link Down_LoopInitializeTimeout
| 00000000  |           0  |Loss of Signal
| 00000000  |           0  |Loss of Received Clock
| 00000000  |           0  |NOS_OLS >200ms continuously Received
| 00000000  |           0  |Link Reset Received
| 00000000  |           0  |LIP F7 Received
| 00000000  |           0  |LIP F8 Received
| 00000000  |           0  |Connected P2P Mode
| 00000000  |           0  |Port Configuration Changed
| 00000000  |           0  |L-Bit Detected
| 00000000  |           0  |Connection Fabric P2P
| 00000000  |           0  |Connection Fabric Loop
| 00000000  |           0  |Connection Private Loop
| 00000000  |           0  |Connection P2P
| 00000000  |           0  |NOS or OLS for >200 mSec (or user defined value) in the
init f/w control block
| 00000000  |           0  |P2P Link Event Timeout
| 00000000  |           0  |Loop Initialize Protocol Error
| 00000000  |           0  |LR Initiated by 16G_CHA Protocol CHIP
| 00000000  |           0  |LRR Received by 16G_CHA Protocol CHIP
| 00000000  |           0  |LIP generated by ISP due to tout when attempt to transmit
non-data frame
| 00000000  |           0  |Response Queue Full
| 00000000  |           0  |ATIO Queue Full
| 00000000  |           0  |Drop AE due to lack of resources
| 00000000  |           0  |ELS Protocol Error Detected by Firmware
| 00000000  |           0  |OPEN Device Failed
| D0B5C113  |   3501572371  |Transmit Frame Count from 16G_CHA Protocol CHIP
| 630B2376  |   1661674358  |Received Frame Count From 16G_CHA Protocol CHIP
| 00000000  |           0  |Discarded Frame Count From 16G_CHA Protocol CHIP
| 00000001  |           1  |Frame Dropped by Firmware
| 00000000  |           0  |LIP Primitives Received
| 00000000  |           0  |NOS Primitive Received
| 00000000  |           0  |OLS Primitive Received
| 00000000  |           0  |Not Used
| 00000000  |           0  |Not Used
```

```
| 00000000  |           0  |Class2 Sequence Timeout
| 00000000  |           0  |P_RJT Frame Tramsmit
| 00000000  |           0  |Failure to allocate exchange resource when receiving a
frame for a new exchange
| 00002630  |        9776  |ABTS Received
| 00000001  |           1  |Received sequences with a missing frame
| 00000000  |           0  |Correctable Error
| 0109FD7E  |    17431934  |Mailbox Commands Issued
| 00000000  |           0  |Failure to allocate NportHandle when receiving an ELS frame
| 00000000  |           0  |Received EOFa
--------------------------------------------------
```

写 I/O 过程及错误特征码（示意）:

WRITE

```
HBA/INI : WRITE CMD(4K)========X0================> TGT/RCU-TGT
                          <Round-Trip Gap-1>
                   <==============X1========== Transfer Ready FRME
              DATA-FRAME-0  =======X2========>
              DATA-FRAME-1  =======X3========>
                          <Round-Trip Gap-2>
                   <==============X4===========STATUS FRAME
```

- On a WRITE CMD Sequence, such as shown above, depending on which FRAME gets lost of damaged, will cause different SSB to be logged and cause a different level of Response Impact to the Host.
- Thus understanding the SSB that are logged and how they correspond with the above errors can help focusing on where to look for errors in a Fabric and/or Internally in a DKC.
- **KEY SSB for WRITE IO related LINK issues : B65C, DDA1, B6FE, D034**
- ***There is some Notes below on when some of above LOG relative to the timing of the "X0 to X4" errors above.***
- The "***Round-Trip Gaps 1 &2***" shown above depend entirely on the DISTANCE between the HBA/INI and Target PORTS. Rule of thumb is "1ms" per "100KM" per Round-Trip. This holds well pretty well if the LINKS are purely DARK-FIBRE LINKS. If there is FCIP in a WAN things can change dramatically to the worse.

2.1.2 应用对应的 RDKC 存储日志分析

从 RDKC 存储（31028）底层日志也可以看到 ZYO32_DB01JQP 应用对应的 4B 端口在部

分时段出现了 I/O 传输延迟的现象，读和写方向都有，

```
2025/03/11 10:24:49  1396   44                : The number of segments per CPK as an additional information of SSB:13F3
2025/03/11 10:24:49  13F3#  44                : s_gfsgm At Hit/Miss, segment reservation timeout occurred (Waiting for free segment)(Inhibited for 1 hour/MPB)
2025/03/11 10:24:29  10F5   4B                : timdiv Async processing was performed forcibly
2025/03/11 10:06:44  FFFB#       4B   PLOGI   : LINK ELS EVENT Type = PLOGI
2025/03/11 10:06:43  B6C0#  3E   4B           : xlmain It FCP Frame-receives from non-LOGIN HOST.
2025/03/11 10:06:43  B358#  02   4B           : qlsnhlliocb LL IOCB was issued
2025/03/11 10:06:43  B3FF   02   4B           : snptovssb Information enhancement for transfer TOV
2025/03/11 10:06:43  B6AD#  02   4B           : tpcc_xlfhostssb Information enhancement for transfer TOV/count error (Host information: Port ID, WWN)
2025/03/11 10:06:43  B6A9#  02   4B           : snptov   Complementing information log for transfer TOV (LM)
2025/03/11 10:06:43  D034#  02   4B   40:6B   : fcmxrwtov Sync command Read/Write JOB TOV
2025/03/11 10:06:43  D031#  02   4B   40:6B   : fcmxssbtv Additional information of SSB=d030/d034 (OPEN command, SCSI reset/TOV)
2025/03/11 10:06:43  DDA1#  02   4B           : cktov_lm Data transfer (Tachyon) completion TOV
2025/03/11 10:06:43  DDA1#  02   4B   40:6B   : cktov_lm Data transfer (Tachyon) completion TOV
2025/03/11 10:06:43  FFFB#       4B   LOGO    : LINK ELS EVENT Type = LOGO
2025/03/11 10:06:34  D555#  02   4B   40:6B   : fcmxd555s Information enhancement log was output because job running time of more than 1 sec was detected
2025/03/11 09:41:46  FFFB#       4B   PLOGI   : LINK ELS EVENT Type = PLOGI
2025/03/11 09:41:45  B6C0#  3D   4B           : xlmain It FCP Frame-receives from non-LOGIN HOST.
2025/03/11 09:41:45  B358#  0C   4B           : qlsnhlliocb LL IOCB was issued
2025/03/11 09:41:45  B3FF   0C   4B           : snptovssb Information enhancement for transfer TOV
2025/03/11 09:41:45  B6AD#  0C   4B           : tpcc_xlfhostssb Information enhancement for transfer TOV/count error (Host information: Port ID, WWN)
2025/03/11 09:41:45  B6A9#  0C   4B           : snptov   Complementing information log for transfer TOV (LM)
```

```
 Date        Time    | SSB   | MP | PORT | LDEV  | Hitachi SSB Refcode Description
2025/03/11 18:22:47   10F5   4F                  : timdiv Async processing was performed forcibly
2025/03/11 17:07:36   10F5   4D                  : timdiv Async processing was performed forcibly
2025/03/11 16:50:24   12DD#  44                  : c_rcvinv  When SCSI was reset, disabled message was received
2025/03/11 16:50:24   16BD   44                  : r_ssvl6bd Target of LtoL MSG clearance was detected
2025/03/11 16:50:24   D031#  44   4B     40:62   : fcmxssbtv Additional information of SSB=d030/d034 (OPEN command, SCSI reset/TOV)
2025/03/11 16:50:24   D030#  44   4B     40:62   : fcmxscrst SCSI reset occurred (Information enhancement)
2025/03/11 16:50:24   16AD#  44   4B             : r_ssbl6al Complementing information 2 of Err=0x1699
2025/03/11 16:50:24   16A1#  44   4B             : sr_jobclr Complementing information of Err=0x1699
2025/03/11 16:50:24   1699#  44   4B             : sr_rstssb SCSI reset processing execution (time etcof reset generating / PSON/Reboot / obstacle.)
2025/03/11 16:41:32   D031#  0E   4B     40:63   : fcmxssbtv Additional information of SSB=d030/d034 (OPEN command, SCSI reset/TOV)
2025/03/11 16:41:32   D030#  0E   4B     40:63   : fcmxscrst SCSI reset occurred (Information enhancement)
2025/03/11 16:41:32   16AD#  0E   4B             : r_ssbl6al Complementing information 2 of Err=0x1699
2025/03/11 16:41:32   16A1#  0E   4B             : sr_jobclr Complementing information of Err=0x1699
2025/03/11 16:41:32   1699#  0E   4B             : sr_rstssb SCSI reset processing execution (time etcof reset generating / PSON/Reboot / obstacle.)
2025/03/11 16:39:53   12DD#  0A                  : c_rcvinv  When SCSI was reset, disabled message was received
2025/03/11 16:39:53   16BD   0A                  : r_ssvl6bd Target of LtoL MSG clearance was detected
2025/03/11 16:39:53   D031#  0A   4B     40:61   : fcmxssbtv Additional information of SSB=d030/d034 (OPEN command, SCSI reset/TOV)
2025/03/11 16:39:53   D030#  0A   4B     40:61   : fcmxscrst SCSI reset occurred (Information enhancement)
2025/03/11 16:39:53   16AD#  0A   4B             : r_ssbl6al Complementing information 2 of Err=0x1699
2025/03/11 16:39:53   16A1#  0A   4B             : sr_jobclr Complementing information of Err=0x1699
2025/03/11 16:39:53   1699#  0A   4B             : sr_rstssb SCSI reset processing execution (time etcof reset generating / PSON/Reboot / obstacle.)
2025/03/11 15:46:18   AFB0   3C                  : rcfbk_flg Config BK during Online completed
```

- 写 I/O 延迟，原因为“等待数据传输”，指向存储外部链路，依然和同一块 FC-HBA 卡有关（WWN=100000620B3AC203）：

DATE	TIME	PORT	LDEV#	CMD	KEY/ASC	HG-NAME	WAIT REASON	Host WWN
2025/03/20	22:53:36	4B	00:40:6C	8A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/20	09:55:16	4B	00:40:6D	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/20	09:51:53	4B	00:40:6D	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/20	09:36:02	4B	00:40:6D	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/18	22:53:28	4B	00:40:6C	8A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/12	10:35:00	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/11	10:06:43	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/11	09:41:45	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/10	14:56:07	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/10	14:42:27	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/03/10	14:03:16	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/02/13	10:47:46	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/02/13	10:18:25	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203

2025/02/13	10:12:59	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/02/13	10:08:54	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/02/13	10:04:38	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203
2025/02/13	09:53:47	4B	00:40:6B	2A	B/C001	ZYO32_DB01JQP	Wait for Data Transfer	100000620B3AC203

- 读 I/O 延迟，出现的频率更高一点，原因主要为“等待数据传输”，指向存储外部链路，依然和同一块 FC-HBA 卡有关（WWN=100000620B3AC203）：

DATE	TIME	PORT	RESET TYPE	WAIT REASON	Possible HG-NAME	Possible HBA-WWN
3/20/2025	22:58:43	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	22:56:06	4B	ABTS(Abort Sequence)	Wait due to Slot Busy	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	22:52:09	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	22:52:08	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	22:49:11	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	22:48:18	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	16:52:27	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	16:48:38	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	16:46:54	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	16:38:16	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	13:01:20	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	9:54:10	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/20/2025	0:52:19	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/19/2025	23:10:53	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	23:14:40	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	23:13:57	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	23:13:32	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	22:59:51	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	22:54:01	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	22:53:46	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:38:28	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:32:06	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:29:43	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:27:20	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:25:59	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:23:10	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:22:29	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:18:10	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:16:38	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	19:15:38	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203

3/18/2025	16:53:15	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/18/2025	16:45:05	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/17/2025	20:05:15	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/13/2025	1:24:46	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/12/2025	20:30:07	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/12/2025	16:57:10	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/12/2025	16:43:21	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	23:45:34	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	23:13:58	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	23:00:34	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	22:58:43	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	22:51:26	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	19:53:33	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	19:50:55	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	19:46:28	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	19:45:45	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	19:37:34	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	16:50:24	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	16:41:32	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/11/2025	16:39:53	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	23:13:20	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	22:48:44	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	22:46:41	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:44:44	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:44:23	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:44:00	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:34:27	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:32:49	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:28:39	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:28:33	4B	ABTS(Abort Sequence)	Wait due to Slot Busy	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:26:05	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:25:06	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:24:58	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:23:49	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:20:55	4B	ABTS(Abort Sequence)	Wait due to Slot Busy	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:20:40	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	19:17:45	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	16:51:52	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	16:50:57	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203

3/10/2025	16:49:43	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
3/10/2025	16:43:13	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:34:38	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:33:20	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:32:36	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:29:28	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:27:23	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:21:24	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:20:56	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:20:28	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:20:26	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:20:17	4B	ABTS(Abort Sequence)	Wait due to Slot Busy	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:18:06	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:15:15	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:12:34	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:11:16	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:08:35	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:08:29	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:08:03	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:03:54	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:03:14	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:02:43	4B	ABTS(Abort Sequence)	Wait for Stage/Destage	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	23:02:31	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/13/2025	11:19:49	4B	ABTS(Abort Sequence)	Wait for DMA Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/12/2025	10:32:58	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/12/2025	0:07:03	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/12/2025	0:03:06	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203
2/11/2025	23:54:35	4B	ABTS(Abort Sequence)	Wait for Data Transfer	ZYO32_DB01JQP	100000620B3AC203

类似地，存储上 4B 端口错误计数器并未出现异常增长，可排除存储 4B 端口 SFP 模块及端口到交换机的链路问题：

```
PORT LABEL = 4B
16G FC Port Statistics
| Hex_Count | Decimal_Count | Port_Stats_Description
|-----------|---------------|----------------------------------------------------
----------------------------------------
| 00000000  |            0  |Link Failure
| 00000000  |            0  |Loss of Sync
```

```
| 00000000  |          0  |Loss of Signal
| 00000000  |          0  |Primitive Sequence Protocol Error
| 00000000  |          0  |Invalid Transmission Word
| 00000000  |          0  |Invalid CRC
| 00000000  |          0  |LIP Occurred
| 00000001  |          1  |Link UP
| 00000000  |          0  |Link Down_LoopInitializeTimeout
| 00000001  |          1  |Loss of Signal
| 00000000  |          0  |Loss of Received Clock
| 00000000  |          0  |NOS_OLS >200ms continuously Received
| 00000000  |          0  |Link Reset Received
| 00000000  |          0  |LIP F7 Received
| 00000000  |          0  |LIP F8 Received
| 00000001  |          1  |Connected P2P Mode
| 00000000  |          0  |Port Configuration Changed
| 00000000  |          0  |L-Bit Detected
| 00000001  |          1  |Connection Fabric P2P
| 00000000  |          0  |Connection Fabric Loop
| 00000000  |          0  |Connection Private Loop
| 00000000  |          0  |Connection P2P
| 00000000  |          0  |NOS or OLS for >200 mSec (or user defined value) in the
init f/w control block
| 00000000  |          0  |P2P Link Event Timeout
| 00000000  |          0  |Loop Initialize Protocol Error
| 00000000  |          0  |LR Initiated by 16G_CHA Protocol CHIP
| 00000000  |          0  |LRR Received by 16G_CHA Protocol CHIP
| 00000000  |          0  |LIP generated by ISP due to tout when attempt to transmit
non-data frame
| 00000000  |          0  |Response Queue Full
| 00000000  |          0  |ATIO Queue Full
| 00000000  |          0  |Drop AE due to lack of resources
| 00000000  |          0  |ELS Protocol Error Detected by Firmware
| 00000000  |          0  |OPEN Device Failed
| 04B96737  |    79259447  |Transmit Frame Count from 16G_CHA Protocol CHIP
| DF924962  |   3750906210  |Received Frame Count From 16G_CHA Protocol CHIP
| 00000000  |          0  |Discarded Frame Count From 16G_CHA Protocol CHIP
| 00000000  |          0  |Frame Dropped by Firmware
| 00000000  |          0  |LIP Primitives Received
| 00000001  |          1  |NOS Primitive Received
| 00000000  |          0  |OLS Primitive Received
| 00000000  |          0  |Not Used
```

```
| 00000000  |            0  |Not Used
| 00000000  |            0  |Class2 Sequence Timeout
| 00000000  |            0  |P_RJT Frame Tramsmit
| 00000000  |            0  |Failure to allocate exchange resource when receiving a
frame for a new exchange
| 00002AED  |        10989  |ABTS Received
| 00000000  |            0  |Received sequences with a missing frame
| 00000000  |            0  |Correctable Error
| 010A5D00  |     17456384  |Mailbox Commands Issued
| 00000000  |            0  |Failure to allocate NportHandle when receiving an ELS frame
| 00000000  |            0  |Received EOFa
```

读 I/O 过程及错误特征码（示意）：

READ

```
HBA/INI : READ CMD(4K)=======X0===============> TGT/RCU-TGT
                 <==X1=====DATA-FRAME-0
                 <=========DATA-FRAME-1
                           <==X2==============STATUS FRAME
```

- READ LINK errors are difficult to track inside a DUMP. If the CMD is received OK, then the TARGET PORT simply sends the DATA and STATUS, without any feedback and has to assume the Frames get delivered OK.
- Likewise if the CMD never gets to the DKC Port (error **X0**) then the DKC PORT sends nothing.
- Should an error happen at timing (**X1**) the HBA/INI will get to know about it and issue an ABTS immediately. If the error is at TIMING **X0** or **X2** then the HBA/INI will **TIMEOUT** and issue an ABTS.

```
HBA/INI : ABTS(XID) =============================> TGT/RCU-TGT
                             <==========ACCEPT
```

Depending on where the ERROR happened the DKC PORT will either LOG :

a) RESETD LOG (RESET Received)

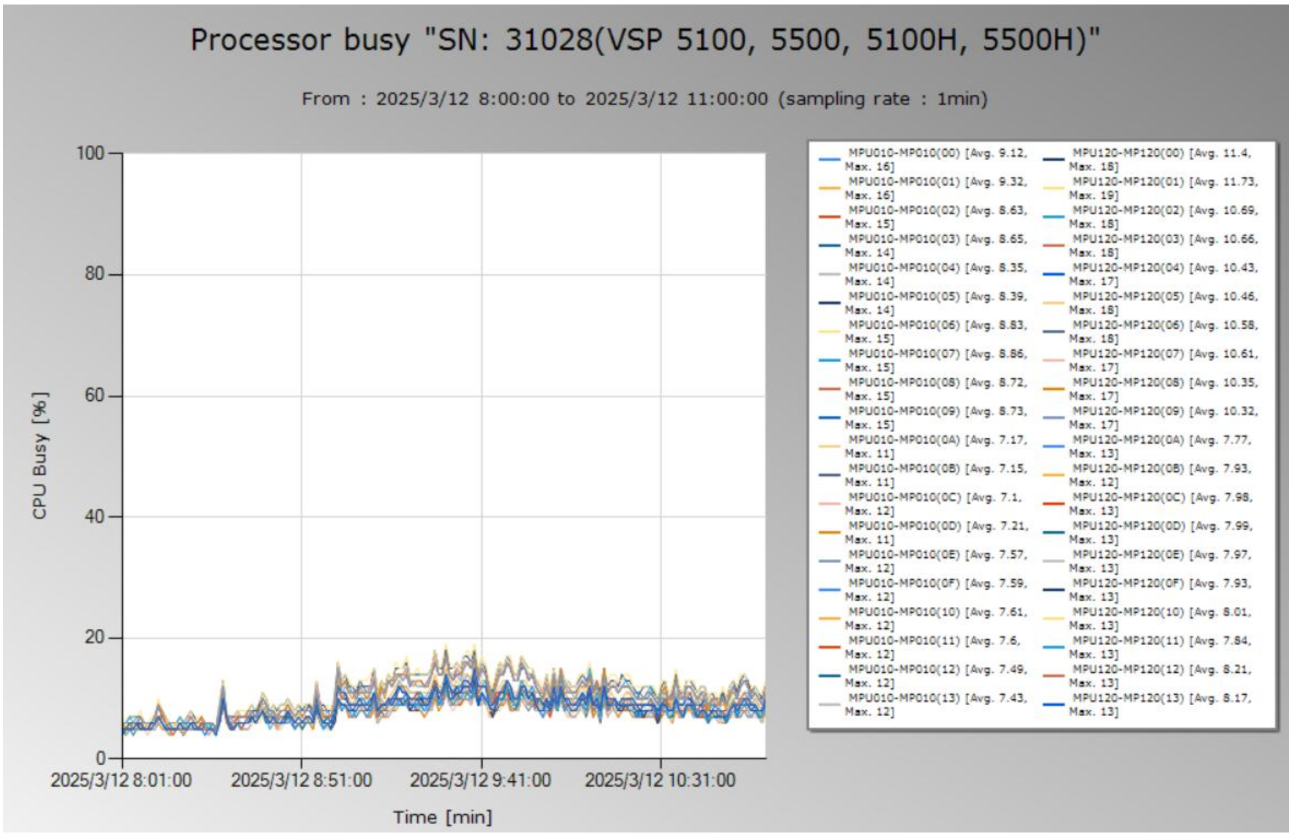
b) SSB = 1699 / 16A1 / 16AD / 16AE (RESET Received)

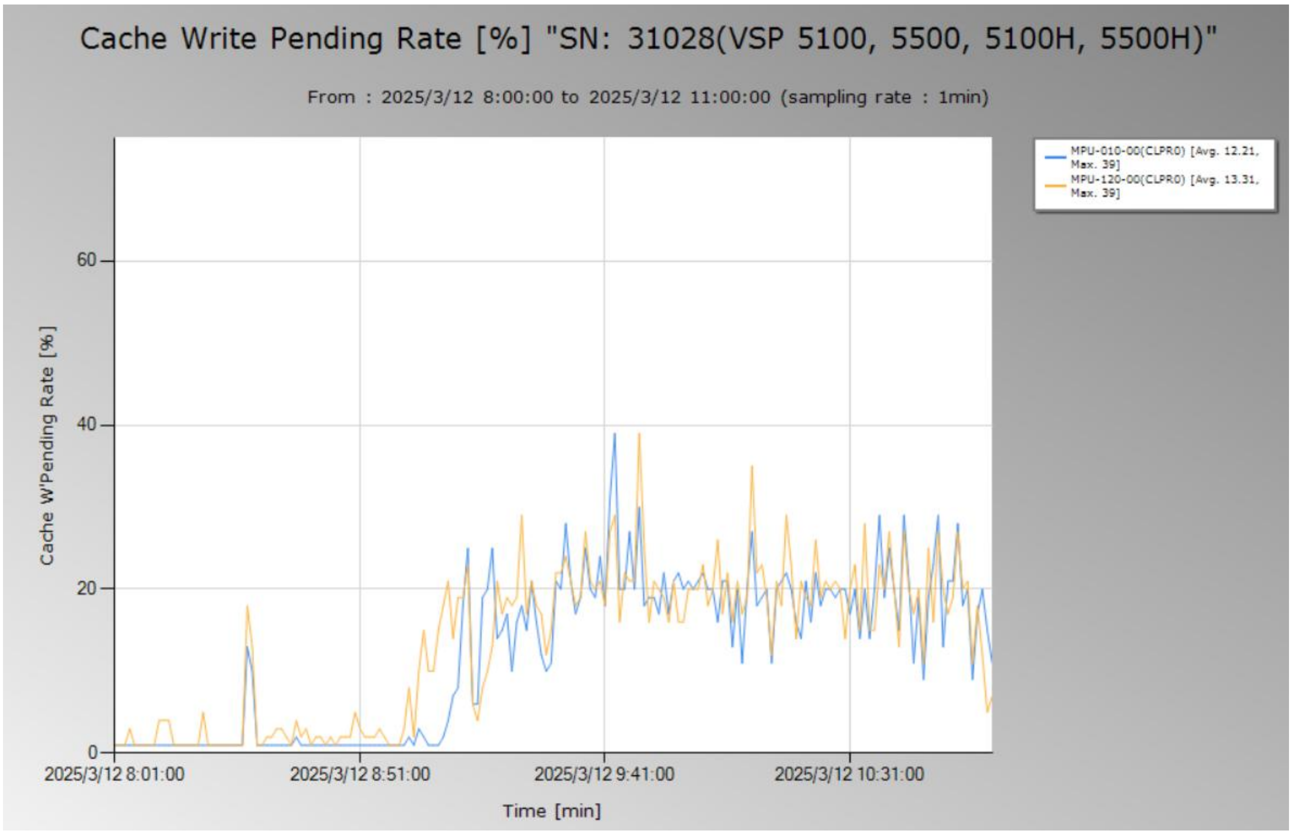
c) SSB = 8907 (ABTS Reset sent after Timeout waiting for Response)

2.2 性能数据分析

2.2.1　存储资源利用率分析

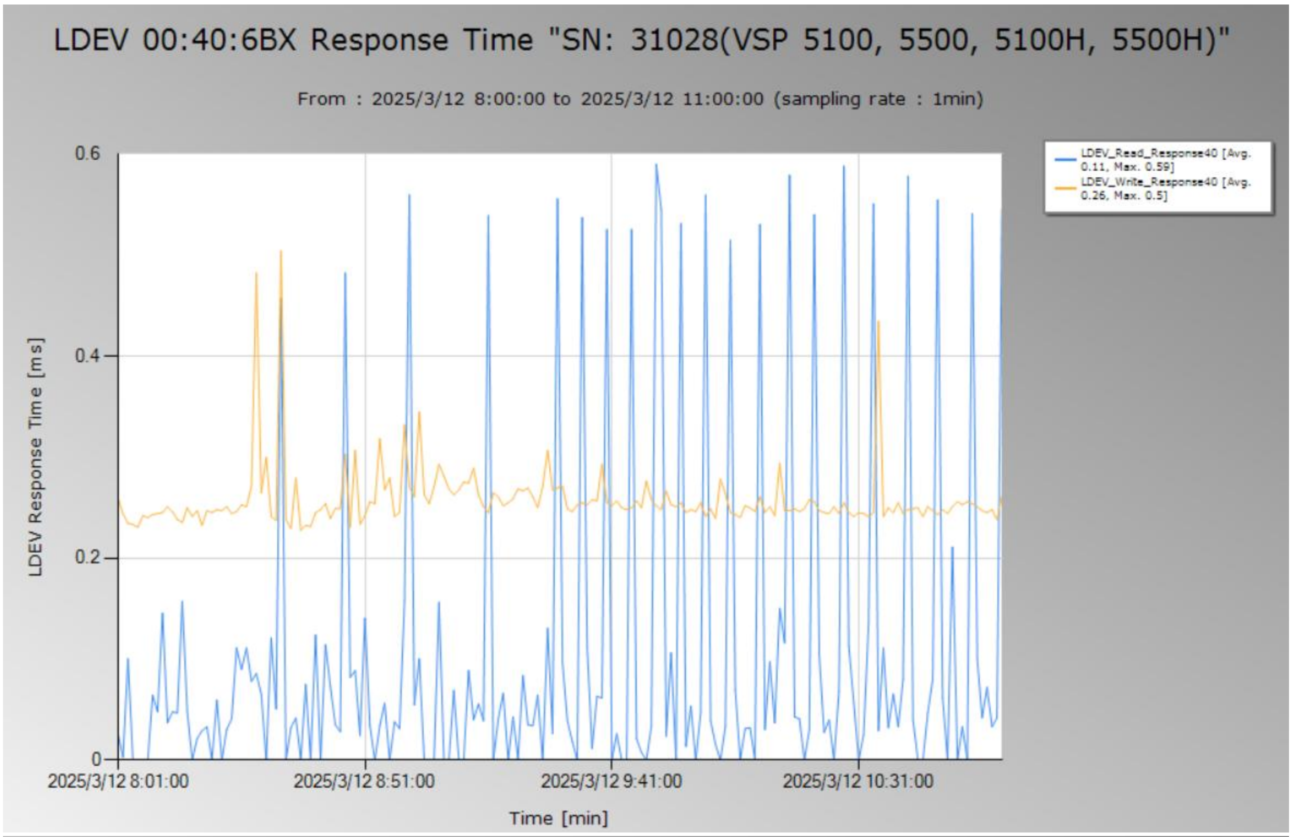
存储全局共享资源 CPU 和 Cache 利用率均在合理范围内，未见异常，说明存储资源未出现瓶颈：

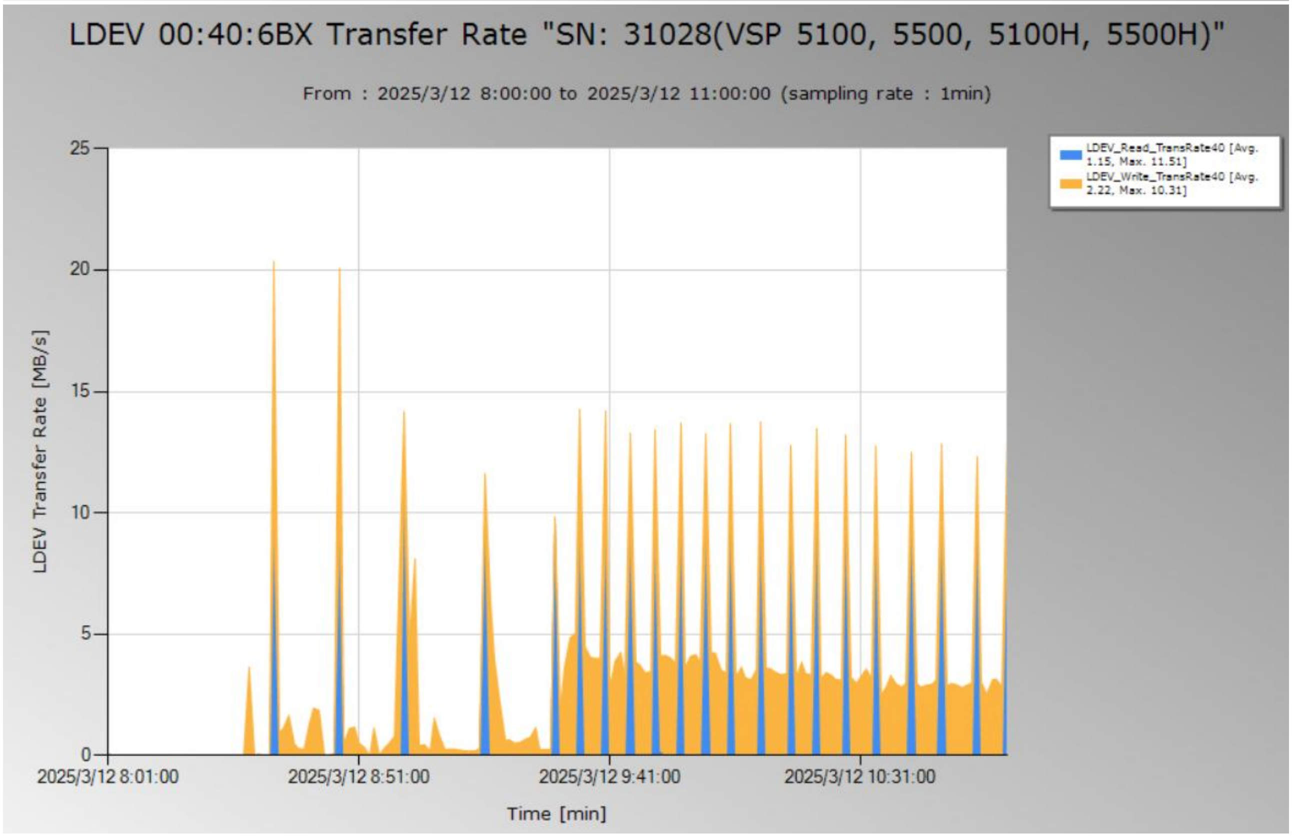


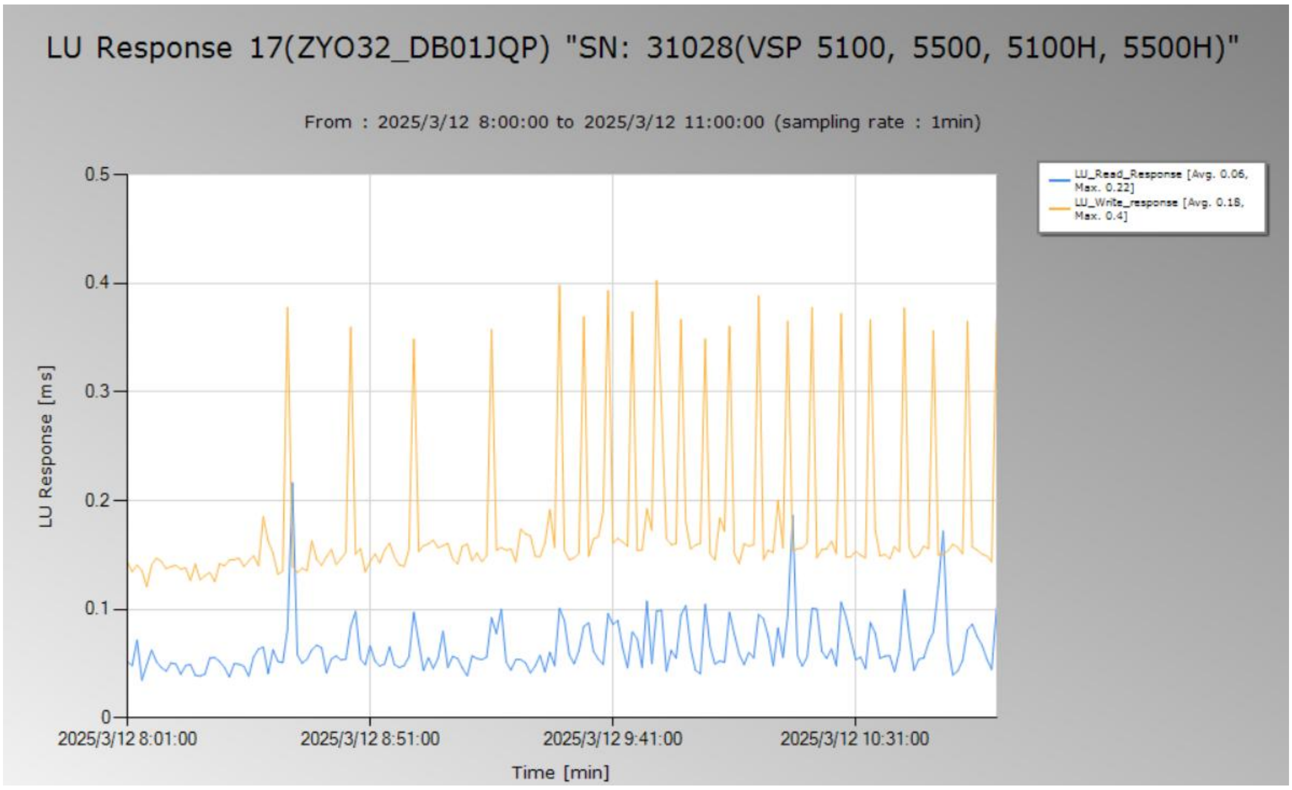


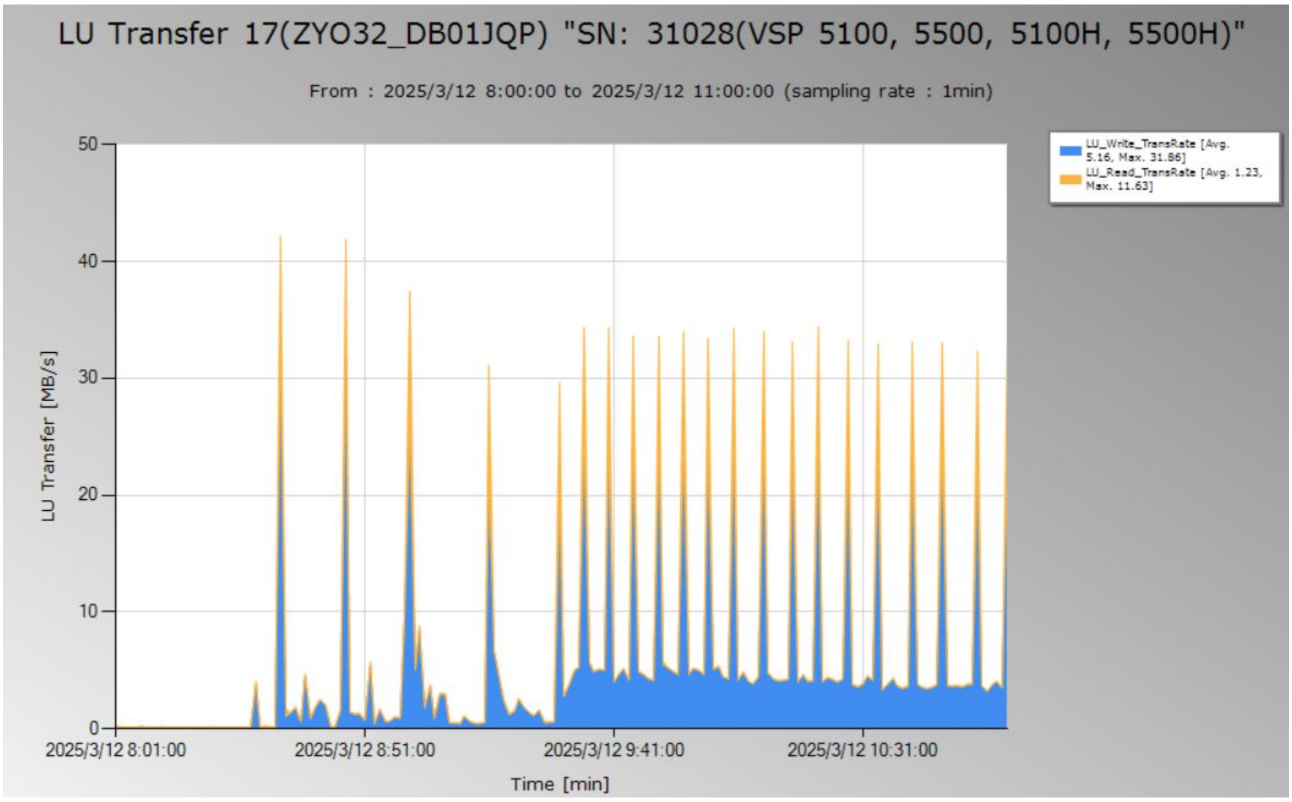
2.2.2 Redo 日志卷性能分析

存储上显示 ZYO32_DB01JQP 应用所使用的 Redo 日志卷 40:6B 读写响应时间良好：写延迟在 0.23ms 左右，读延迟在 0.1~0.5ms 之间波动（波动和负载及链路有关），存储上观测到的性能压力较小，不存在性能问题：


LDEV 00:40:6BX Response Time "SN: 31028(VSP 5100, 5500, 5100H, 5500H)"
From : 2025/3/12 8:00:00 to 2025/3/12 11:00:00 (sampling rate : 1min)
LDEV_Read_Response40 [Avg. 0.11, Max. 0.59]
LDEV_Write_Response40 [Avg. 0.26, Max. 0.5]
LDEV Response Time [ms]
0.6
0.4
0.2
0
2025/3/12 8:01:00
2025/3/12 8:51:00
2025/3/12 9:41:00
2025/3/12 10:31:00
Time [min]


LDEV 00:40:6BX Transfer Rate "SN: 31028(VSP 5100, 5500, 5100H, 5500H)"
From : 2025/3/12 8:00:00 to 2025/3/12 11:00:00 (sampling rate : 1min)
LDEV_Read_TransRate40 [Avg. 1.15, Max. 11.51]
LDEV_Write_TransRate40 [Avg. 2.22, Max. 10.31]
LDEV Transfer Rate [MB/s]
25
20
15
10
5
0
2025/3/12 8:01:00
2025/3/12 8:51:00
2025/3/12 9:41:00
2025/3/12 10:31:00
Time [min]

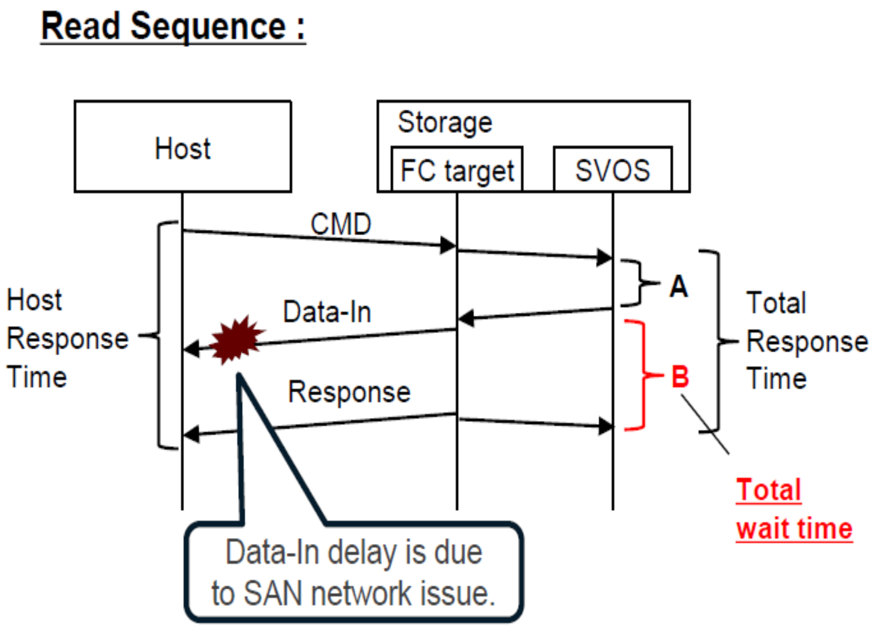

LU Response 17(ZYO32_DB01JQP) "SN: 31028(VSP 5100, 5500, 5100H, 5500H)"
From : 2025/3/12 8:00:00 to 2025/3/12 11:00:00 (sampling rate : 1min)
LU_Read_Response [Avg. 0.06, Max. 0.22]
LU_Write_response [Avg. 0.18, Max. 0.4]
LU Response [ms]
0.5
0.4
0.3
0.2
0.1
0
2025/3/12 8:01:00
2025/3/12 8:51:00
2025/3/12 9:41:00
2025/3/12 10:31:00
Time [min]


LU Transfer 17(ZYO32_DB01JQP) "SN: 31028(VSP 5100, 5500, 5100H, 5500H)"
From : 2025/3/12 8:00:00 to 2025/3/12 11:00:00 (sampling rate : 1min)
LU_Write_TransRate [Avg. 5.16, Max. 31.86]
LU_Read_TransRate [Avg. 1.23, Max. 11.63]
LU Transfer [MB/s]
50
40
30
20
10
0
2025/3/12 8:01:00
2025/3/12 8:51:00
2025/3/12 9:41:00
2025/3/12 10:31:00
Time [min]

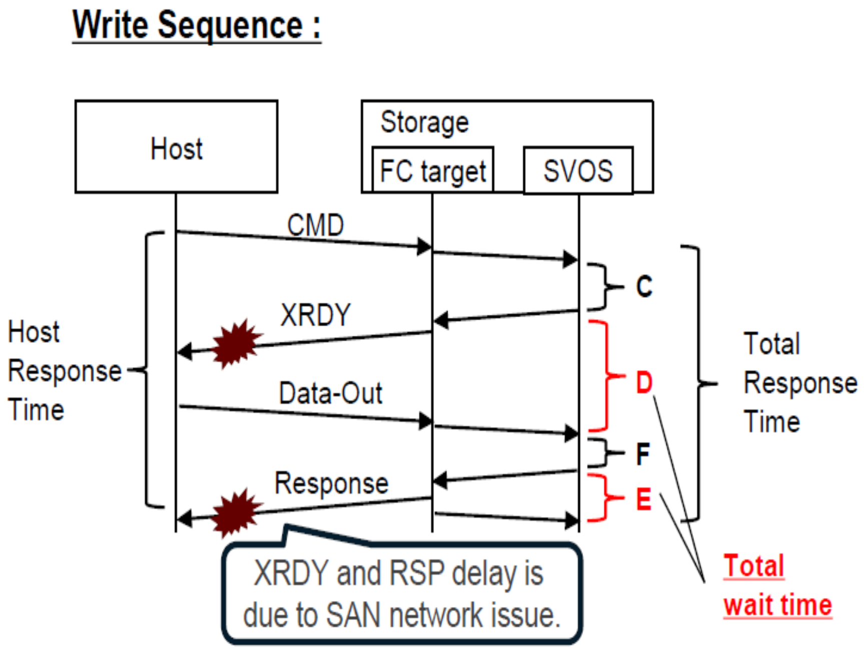
2.2.3 关于存储读写响应时间

存储上的响应时间从数据帧被传输到存储端口开始计算，因此主机上看到的响应时间和存储端不会完全一致，如果存储上的响应时间处于正常范围内，而主机上的响应时间偏大，就可能是由于中间链路传输耗时所导致。

读 I/O：延时可能出现在将目标数据传输至主机时所经过的链路



写 I/O：延时可能出现在 XRDY 状态帧传输至主机时所经过的链路以及和主机确认写完成信号时所经过的链路



3 结论和建议

综上分析，Oracle 数据库应用的 Redo 日志卷出现时延增加是由于存储外部问题，主要指向 FC-HBA 光纤卡（WWN=100000620B3AC203）到交换机这一侧的链路，建议重点排查这一段的链路及所有相关组件（FC-HBA、光纤线及 SFP 模块）尤其是 FC-HBA 卡，或者通过隔离法排除怀疑部件。

如果此类问题在隔离和排查后依然存在，建议如下：

1) 针对 Redo 日志卷收集秒级性能数据（由于秒级收集的收集会产生性能开销，不建议持续收集，一般收集 10 分钟左右，可结合定时任务分段收集）；
2) 根据上述分析，存储的响应时间受到所经过的传输链路影响，如果怀疑存储内部资源处理 I/O 耗时过久导致应用延迟变高，可部署收集程序获取存储内部处理 I/O 用时